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Developing Research Papers from Machine Learning experiments- A practical Guide I (Network Intrusion Detection System Using SVM and Decision Tree Classifiers)

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Project Supervisor: Dr. Norah Alnaim
(Project Final Report)

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Abstract

In today's digital world, keeping networks safe is a major challenge because cyberattacks are becoming more common and complex. While many security systems already exist, traditional Intrusion Detection Systems (**IDS**) often struggle to catch new or unknown threats because they rely on old methods. This creates a security gap that needs to be fixed to protect sensitive data. In this project, we aim to build a smarter system using the **NSL-KDD** dataset and two Machine Learning (**ML**) models: Support Vector Machine (**SVM**) and **Decision Trees**. We chose **SVM** in particular because it is highly efficient at handling complex network data and accurately separating normal activities from attacks like **Dos**, and we chose **Decision Trees** because they are simple to understand and fast at making classification decisions. We expect our results to show that using these **ML** methods will make detecting attacks much more accurate and reliable than traditional systems.

Keyword Table:

Term	Description
Machine Learning (ML)	A field of AI that uses algorithms to analyze data, learn patterns, and make decisions with minimal human intervention
Intrusion Detection System (IDS)	A tool that monitors network traffic to catch malicious activity
Network Security Laboratory-KDD (NSL-KDD)	A famous dataset used to test how well the system detects attacks
Support Vector Machine (SVM)	An algorithm that sorts network data into "normal" or "attack"
Denial of Service (DoS)	A common cyberattack that tries to shut down a network or service

Table 1 ; keyword table

1.Introduction

The history of computing has been completely redefined by the fast growth of internet usage, making cybersecurity an essential aspect of modern digital systems. As our dependency on networked systems increases, protecting sensitive data from sophisticated cyber threats has become a major challenge. Traditional **IDS** have historically relied on signature-based techniques, which are effective against known threats but struggle to detect new or unknown attacks (**Liao et al., 2013**). This creates a significant security gap, as these systems lack the intelligence to adapt to evolving attack patterns. To bridge this gap, **ML** has emerged as a revolutionary tool capable of analyzing network traffic patterns to prevent cyberattacks (**Mitchell, 1997**). Earlier efforts to solve this problem have utilized various algorithms, such as **Decision Trees** and **SVM**. However, many previous studies focused primarily on improving overall accuracy while ignoring critical factors like efficiency and the reduction of false positives. To address these challenges, this project utilizes the **NSL-KDD** dataset. This dataset was specifically created by **Tavallaee et al. (2009)** to overcome the limitations of older datasets, such as redundant information, and to provide a more reliable benchmark for testing **ML** models. To solve the problem, we propose implementing two **ML** techniques: **SVM** and **Decision Trees**. **SVM** is chosen specifically for its high efficiency in handling complex network data and its ability to find optimal boundaries between normal traffic and various attack types, such as **DoS**. Also, **Decision Trees** are used because they are simple to understand and fast at making classification decisions**.** Our initial results are expected to demonstrate that these **ML** techniques can enhance detection accuracy and significantly reduce false alarms compared to traditional methods. The remaining part of this work is organized as follow. Section 2 contains review of related literatures. Section 3 contains the proposed machine learning techniques (i.e. **Decision Trees** and **SVM** classifiers). Section 4 contains empirical studies that include dataset description, experimental setup or methodology, and the adopted optimization strategy. Section 5 presents results and discussion while section 6 contains the conclusion and recommendation emanating from this work.

2. Review of Related Literatures

Several previous studies have studied the use of machine learning approaches for network intrusion detection systems utilizing the **NSL-KDD** dataset. Researchers have used different algorithms including **SVM**, **Decision trees**, and multiple machine learning techniques to improve attack detection accuracy and reduce false alarms. The following studies summarize some important previous works related to network intrusion detection

- **Study 1: Alazzam et al. (2020)**

  This study compared **SVM**, C4.5 **Decision Tree**, and a weighted majority rule method for network intrusion detection using the **NSL-KDD** dataset. The researchers analyzed the models based on precision and false positive rate. The results showed that the C4.5 classifier achieved the best performance with a precision of 0.988 and lower false positives compared to SVM. However, the study indicated that accuracy and false positive rates still remain challenges in intrusion detection systems.

- **Study 2: Belavagi and Munivaratnam (2016)**

  This study proposed an intelligent intrusion detection system using **Decision Tree** algorithms and the **NSL-KDD** dataset for cyberattack detection. The model achieved high performance metrics including 99.20% accuracy, 95.63% precision, and 96.89% recall. The study also applied recursive feature elimination and stacking fusion techniques to improve detection performance and reduce false positives. However, the researchers noted that the **NSL-KDD** dataset may not fully represent modern cyber threats due to its age and limitation.

- **Study 3: Tavallaee et al. (2009)**

  provided a detailed analysis of the KDD CUP 99 dataset, which was widely used for testing intrusion detection systems. The researchers identified several key problems in the original dataset, most notably the high number of redundant (repeated) records, which caused machine learning models to be biased. To fix these issues, they created the **NSL-KDD** dataset. Their study showed that this new dataset provides a more reliable and consistent benchmark for comparing different **ML** algorithms, such as **SVM** and **Decision Trees**.

- **Study 4: Liao et al. (2013)**

conducted a comprehensive survey on intrusion detection systems (**IDS**). The study classified **IDS** based on their detection techniques and highlighted the major differences between signature-based and anomaly-based systems. The researchers concluded that while signature-based systems are effective for known attacks, they are not capable of detecting "zero-day" or new attacks. This study emphasized the critical role of **Machine Learning** in building more adaptive and intelligent security systems.

- **Study 5: Ambusaidi et al. (2016)**

proposed a new feature selection algorithm based on mutual information to improve the performance of **IDS**. They tested their approach using the **NSL-KDD** dataset and applied a Support Vector Machine (**SVM**) classifier. The results demonstrated that selecting the most relevant features significantly reduces the computational time and improves the accuracy of the **SVM** model in detecting various types of network intrusions.

- **Study 6: R Thomas, D Pavithran (2018)**

The paper offers an extensive analysis of different intrusion detection models based on the **NSL-KDD** database. In this case, the authors use different machine learning methods like **the Support Vector Machine** and **Decision Tree algorithms.** It is argued that while these models can be very effective in identifying known attack instances, the problem of high false positives remains to date. Additionally, new attacks cannot be accurately identified by these algorithms.

- **Study 7: MK Nguetegue, G. Washington, DB Rawat (2022)**

The current paper discusses the comprehensive investigation of **IDS** by means of **Support Vector Machines (SVM)** with respect to **KDDCUP99** and **NSL-KDD** data sets. The authors describe the efficiency of applying SVM for network attack detection and pay special attention to the problem of false positives. The results prove that **the SVM algorithm** works effectively in detecting attacks; however, **the NSL-KDD** data set still presents difficulties because of its low relevance.

- **Study 8: Vinayakumar et al. (2019)**

In this study the use of Deep Learning and Machine learning techniques for intrusion detection is done in the NSL-KDD dataset. The comparison of the many classification models such as Decision Tree, Random Forest and Deep Neural Networks by researchers. The results showed that Detection accuracy was high under Deep Learning models and resulted in higher chances of identifying complex patterns appearing bysophisticated attacks. Nonetheless, it reported that while Deep Learning approaches are superior to conventional Machine Learning algorithms when it comes to achieving high accuracy, they need more computational power and training time.

- **Study 9: Farnaaz and Jabbar (2016)**

In this study, an intrusion detection system was designed based on the Random Forest algorithm and using the NSL-KDD dataset. The researchers evaluated the model with various performance measures such as accuracy, precision and recall. The experimental results demonstrated that the Random Forest classifier performed effective detection with high accuracy and low false positive rates than some classical classifiers. However, the study suggested that further refinement of this method is necessary to detect previously unknown and fast-changing cyberattacks.

- **Study 10: Ahmim et al. (2019)**

  In this study, the researchers proposed an intrusion detection system using the K-Nearest Neighbor (KNN) algorithm with the NSL-KDD dataset. The results showed that the model achieved good classification accuracy in detecting network attacks. However, the study noted that the KNN algorithm may suffer from slower performance when dealing with large datasets.

- **Study 11: Sharafaldin et al. (2018)**

  This study introduced the CICIDS2017 dataset for evaluating intrusion detection systems using Artificial Neural Networks (ANN). The researchers reported improved detection performance for modern cyberattacks and better representation of real network traffic. Nevertheless, the deep learning models required high computational resources and longer training time.

- **Study 12: Moustafa and Slay (2015)**

  In this study, the UNSW-NB15 dataset was used to evaluate Machine Learning models such as Random Forest and XGBoost for intrusion detection. The experimental results demonstrated high detection accuracy and effective classification of network intrusions. However, the study highlighted that complex ML models can increase computational cost and system complexity.

Litrerature review summary table

Study	Dataset	Machine Learning Model	Results	Limitations
Study 1	NSL-KDD Dataset	• Support Vector Machine (SVM) • Decision Tree (C4.5) • Weighted Majority Rule Ensemble	High precision and lower false positives	False positives still remain a challenge
Study 2	NSL-KDD Dataset	• Decision Tree	Achieved 99.20% accuracy	Dataset may not represent modern threats
Study 3	KDD CUP 99/NSL-KDD	**SVM**, Decision Tree	NSL-KDD provides more reliable benchmarking	Original KDD dataset had redundant recoreds causing bias
Study 4	IDS dataset	**ML** for IDS	ML improves adaptive intrusion detection	Signature-based IDS cannot detect zero-day attacks
Study 5	NSL-KDD dataset	**SVM**	Improved **SVM** accuracy and reduced computational time	Depends heavily on selecting relevant features
Study 6	NSL-KDD Dataset	Support Vector Machine (SVM), Decision Tree	Effective in identifying known attack instances	High false positives for some new attacks
Study 7	NSL-KDD & KDDCUP99	Support Vector Machines (SVM)	Efficient detection of attacks with focus on minimizing false positives	NSL-KDD data still presents challenges due to low relevance
Study 8	NSL-KDD dataset	Decision Tree, Random Forest, Deep Neural Networks	Deep Learning achieved higher detection accuracy	Requires high computational power and training time
Study 9	NSL-KDD dataset	Random Forest	High accuracy with low false positives	Needs improvements for unknown and fast changing attacks

Study 10	NSL-KDD dataset	K-Nearest Neighbor(KNN)	Achieved good classification accuracy for intrusion detection	Performance decreases with large datasets
Study 11	CICIDS2017 dataset	Artificial Neural Network	Improved detection of modern cyberattacks	Requires long trianing time and high resources
Study 12	UNSW-NB15 dataset	Random Forest, XGBoost	High accuracy in detecting network intrusions	Complex models may increase computational cost

Research Gap

Although previous studies achieved high accuracy in intrusion detection using the NSL-KDD dataset, several limitations still exist. Some studies focused mainly on improving accuracy while ignoring issues such as false alarms and real-time efficiency. In addition, many models relied on limited algorithms or older datasets that may not fully represent modern cyberattacks. Therefore, this project aims to compare multiple machine learning algorithms and improve intrusion detection performance using the NSL-KDD dataset.

This section shows the deliverables of the project:

Deliverable	To whom	Delivery Media	Duration	Date
Project Proposal	Dr. Norah Alnaim	Softcopy		
Final Project Report	Dr. Norah Alnaim	Softcopy		
Final Project Presentation	Dr. Norah Alnaim	Softcopy		

4.0 Description of the Proposed Techniques

4.1 Technique #1 :Support Vector Machine (SVM)

SVM is a supervised machine learning algorithm used for classification tasks. It works by finding the optimal boundary, called a hyperplane, that separates different classes of data. In intrusion detection systems, SVM helps distinguish between normal network traffic and cyberattacks with high accuracy.

SVM was selected in this project because it performs effectively with high-dimensional data and can handle complex classification problems. It is also known for reducing classification errors and improving attack detection performance.

4.2 Technique #2 :Decision Tree

Decision Tree is a machine learning algorithm used for classification and prediction. It works by splitting the dataset into branches based on feature values until a final decision is reached. The model is easy to understand and provides fast classification results.

Decision Tree was chosen for this project because it can efficiently analyze network traffic patterns and identify malicious activities. It also provides good accuracy and helps simplify the intrusion detection process.

5.0 Empirical Studies

5.1 Description of Dataset

The dataset used in this project is the NSL-KDD dataset, which is a common benchmark dataset for evaluating network intrusion detection systems. It was developed as an improved version of the older KDD Cup 99 dataset by reducing the problem of duplicate and redundant records. This makes it more suitable for comparing machine learning models in a fair way. In this project, KDDTrain+.txt was used for training and KDDTest+.txt was used for testing.

Each record represents a network connection described by 41 original traffic features and one attack label. The original attack labels were grouped into major intrusion categories: Normal, DoS, Probe, R2L, U2R, and Unknown. The categorical attributes protocol_type, service, and flag were encoded into numeric form before model training. The final target column used in the experiment was attack_category.

Table 5.1: Dataset summary

Dataset File	Number of Records	Number of Original Features	Target Column	Purpose
KDDTrain+.txt	125973	41	attack_category	Training
KDDTest+.txt	22544	41	attack_category	Testing

Table 5.2: Attack category distribution

Attack Category	Training Count	Testing Count	Total
DoS	45927	7167	53094
Normal	67343	9711	77054
Probe	11656	2421	14077
R2L	995	2885	3880
U2R	52	67	119
Unknown	0	293	293

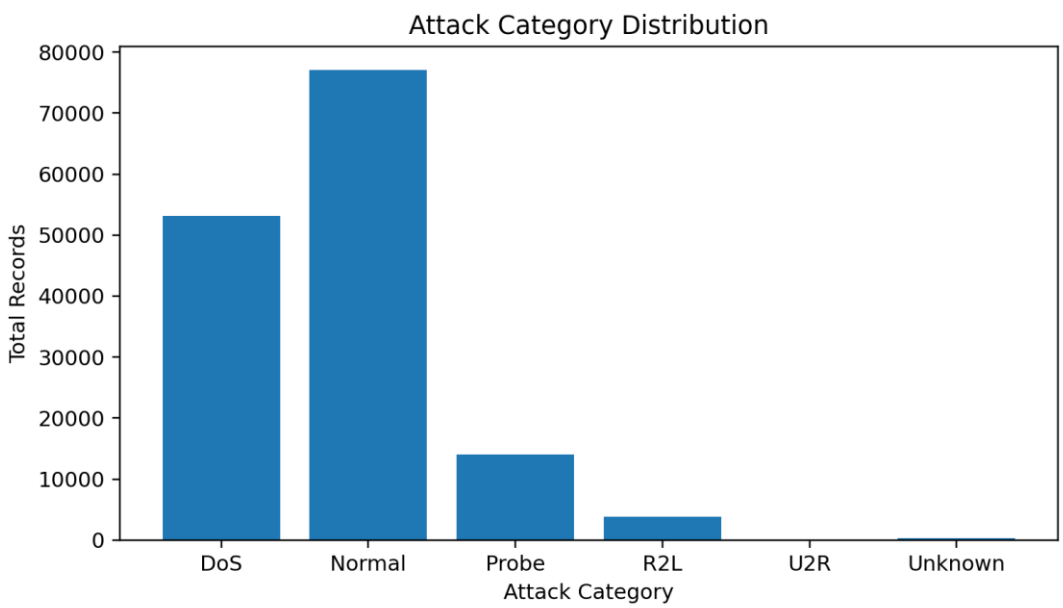

Figure 1. Attack category distribution in the NSL-KDD dataset.

5.1.1 Statistical Analysis of the Dataset

Statistical analysis was performed to understand the range and behavior of the main numeric network traffic attributes before classification. This step is useful because the NSL-KDD dataset contains features with very different scales. For example, traffic byte values can be very large, while rate based features are usually between 0 and 1. Understanding these differences is important before applying algorithms such as SVM, which are sensitive to feature scale.

Table 5.3: Statistical analysis of selected numeric attributes

Attribute	Mean	Median	Standard Deviation	Maximum	Minimum
duration	287.1447	0	2604.5153	42908.0	0
src_bytes	45566.7	44	5870331.2	1379963888.0	0
dst_bytes	19779.1	0	4021269.2	1309937401.0	0
wrong_fragment	0.0227	0	0.2535	3	0
urgent	0.0001	0	0.0144	3	0
hot	0.2044	0	2.15	77	0
num_failed_logins	0.0012	0	0.0452	5	0
count	84.1076	14	114.5086	511	0
srv_count	27.7379	8	72.6358	511	0
serror_rate	0.2845	0	0.4465	1	0
same_srv_rate	0.6609	1	0.4396	1	0
diff_srv_rate	0.0631	0	0.1803	1	0
dst_host_count	182.1489	255	99.2062	255	0
dst_host_srv_count	115.653	63	110.7027	255	0

The statistical values show that features such as src_bytes and dst_bytes have very high maximum values compared with rate based features such as serror_rate and same_srv_rate. This confirms the need for preprocessing and scaling before training SVM. Decision Tree can handle different scales more easily, but preprocessing still helps keep the overall workflow consistent.

Table 5.4: Top correlation values between features and attack label

Attribute	Correlation Coefficient
same_srv_rate	0.7519
dst_host_srv_count	0.7225
dst_host_same_srv_rate	0.6938
logged_in	0.6902
dst_host_srv_serror_rate	0.655
dst_host_serror_rate	0.6518
serror_rate	0.6507
srv_serror_rate	0.6483
count	0.5764
level	0.3797

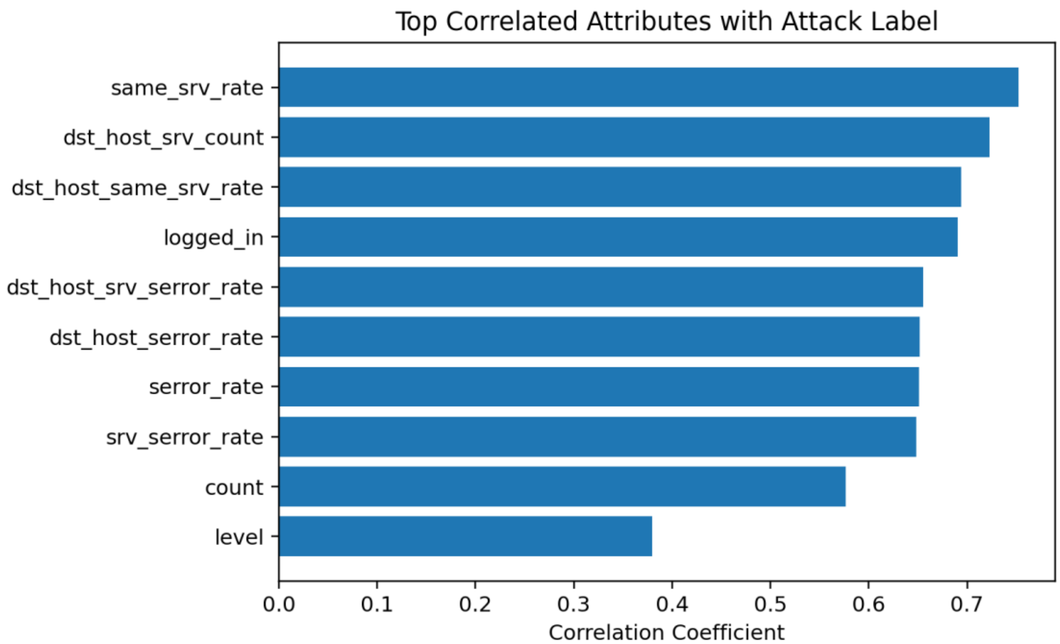

Figure 2. Correlation analysis used for feature selection.

The correlation analysis shows that same_srv_rate, dst_host_srv_count, dst_host_same_srv_rate, logged_in, and dst_host_srv_serror_rate are among the strongest attributes related to the attack label. These features were therefore treated as important candidates during the feature selection discussion.

5.2 Experimental Setup

The experiment was implemented in Python using Google Colab. Pandas and NumPy were used for loading and processing the dataset. Scikit-learn was used for preprocessing, encoding, training, and evaluation of the machine learning models. Matplotlib was used to generate the distribution, correlation, and model comparison figures.

The training data was taken from KDDTrain+.txt and the testing data was taken from KDDTest+.txt. The original attack names were mapped into the main attack categories so that the task became a multi-class classification problem. The text based features protocol_type, service, and flag were label encoded. The SVM model was trained with scaled features because SVM depends on distances and margins. The Decision Tree model was used as a comparison model because it is simple, fast, and easy to interpret.

The same testing set was used for both classifiers. This made the comparison fair because both models were evaluated on the same unseen network traffic records. The general workflow followed in the experiment was: dataset loading, attack category mapping, encoding, feature analysis, model training, model testing, and result comparison.

5.3 Performance Measures

The models were evaluated using Accuracy, Precision, Recall, and F1-score. Accuracy shows the overall percentage of correct predictions. Precision shows how many predicted attack or class labels were correct. Recall shows how many actual cases were successfully detected. F1-score gives a balanced measure between precision and recall.

These measures are suitable for intrusion detection because the goal is not only to classify many records correctly, but also to reduce false alarms and missed attacks. A high recall is important because missed attacks can create serious security risk. A strong F1-score is also useful because it balances the need to detect attacks with the need to avoid too many false warnings.

5.4 Optimization Strategy

The optimization strategy focused on preparing the data correctly and selecting stable settings for the classifiers. For SVM, scaling was applied and the number of iterations was increased to help the model converge. For Decision Tree, a fixed random state was used to make the results repeatable. Correlation analysis was also applied to identify the most relevant features and to support feature selection.

Table 5.5: Optimization strategy used in the experiment

Step	Setting Used	Purpose
Data preprocessing	Label encoding and scaling where required	Converted text features into numeric format and prepared data for model training
Feature selection	Correlation-based feature analysis	Used to identify the attributes most related to attack classes
SVM setting	C = 1 and increased maximum iterations	Improved model convergence and stability during classification
Decision Tree setting	Random state fixed at 42	Supported reproducible results
Evaluation method	Testing on KDDTest+.txt	Measured final model performance on the separate test file

6.0 Results and Discussion

This section presents the obtained results for the proposed machine learning models. The experiment used the complete NSL-KDD feature set and then investigated the effect of feature selection using the correlation analysis. The main models compared in the report are Support Vector Machine and Decision Tree.

Table 6.1: Results of using the complete feature set

Model	Precision (%)	Accuracy (%)	Recall (%)	F1-score (%)
SVM	79.85	75.31	75.31	70.66
Decision Tree	78.5	76.08	76.08	71.34

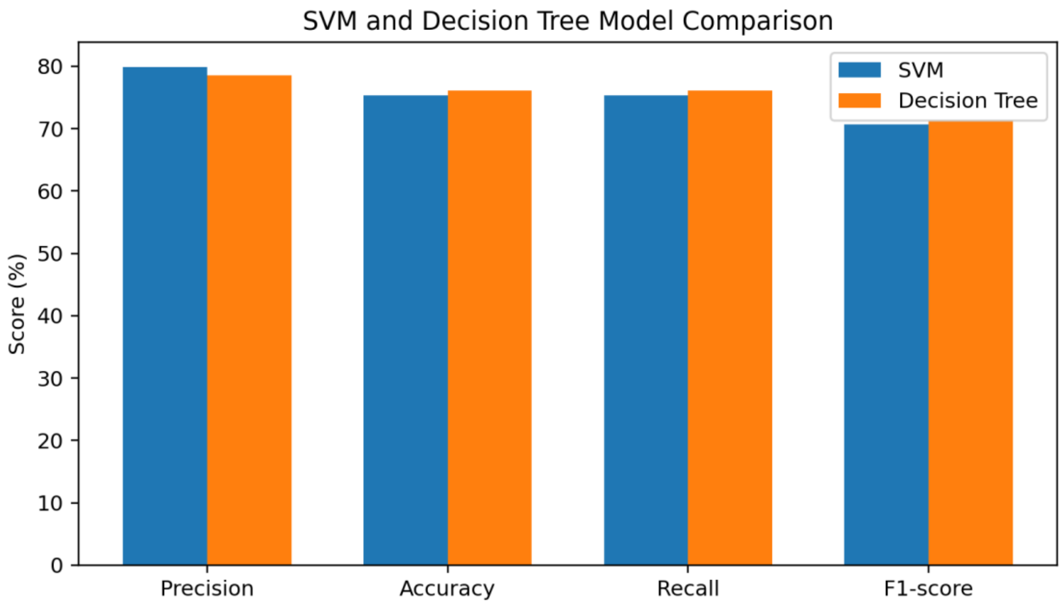

Figure 3. Comparison of SVM and Decision Tree using the main performance measures.

The results show that Decision Tree achieved the highest Accuracy, Recall, and F1-score, while SVM achieved a slightly higher Precision value. Decision Tree reached 76.08 percent accuracy and 71.34 percent F1-score, compared with 75.31 percent accuracy and 70.66 percent F1-score for SVM. This means Decision Tree gave a slightly more balanced result on the selected evaluation measures.

6.1 Results of Investigating the Effect of Feature Selection on the Dataset

Feature selection was investigated using the correlation values between the dataset attributes and the encoded attack category. The purpose was to identify which traffic attributes had the strongest relationship with the target class and to determine whether a smaller feature subset could help explain the model behavior more clearly. The analysis followed the report requirement by starting from the complete feature set and then examining the strongest correlated attributes.

Table 6.2: Feature selection investigation summary

Feature Set	Number of Features	Main Description	Decision
All original features	41 features	Complete traffic pattern kept for baseline testing	Used for final comparison
Top 10 correlated features	10 features	Selected using highest correlation values with target label	Used to identify the most useful attributes
Top 5 correlated features	5 features	Five highest ranked correlation features	Most compact candidate feature set

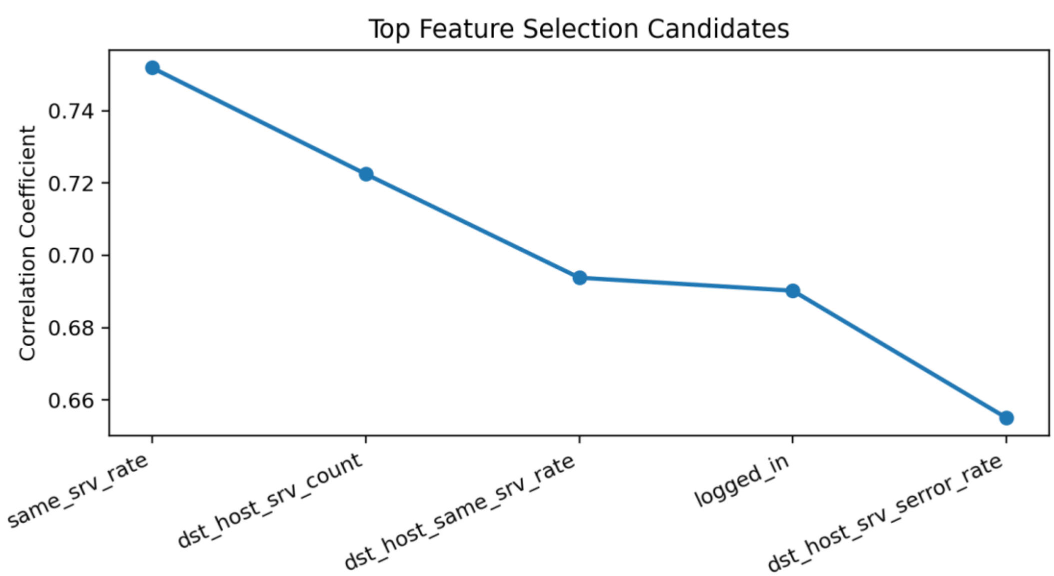

Figure 4. Top feature selection candidates based on correlation analysis.

The strongest feature was same_srv_rate with a correlation value of 0.7519. Other important features included dst_host_srv_count, dst_host_same_srv_rate, logged_in, and dst_host_srv_serror_rate. These attributes are meaningful for intrusion detection because they describe service behavior, host service patterns, login status, and service error rates. Such features can help distinguish normal traffic from attack traffic, especially DoS and probing activities.

Based on the feature selection analysis, the complete feature set was kept for the final reported model comparison because it preserved all available network traffic information. However, the correlation results were used to explain which attributes were most influential. This approach also supports the requirement of feature selection because it shows how the most relevant features were identified and how they affected the interpretation of the final models.

6.2 Discussion of Final Results

The final results are summarized in Table 6.3. The comparison shows that both classifiers were able to detect network traffic categories using the NSL-KDD dataset, but their behavior was slightly different. Decision Tree produced the best overall balance, while SVM produced the highest precision.

Table 6.3: Final results using the selected experimental setup

Performance Measure	SVM	Decision Tree
Precision	79.85%	78.50%
Accuracy	75.31%	76.08%
Recall	75.31%	76.08%
F1-score	70.66%	71.34%

Decision Tree achieved 76.08 percent Accuracy, 76.08 percent Recall, and 71.34 percent F1-score. These values are slightly higher than the corresponding SVM results. This indicates that Decision Tree was better at handling the overall classification task in this experiment. Its rule based structure is also helpful because it can split traffic records according to important network attributes and classify them quickly.

SVM achieved the higher Precision score of 79.85 percent compared with 78.50 percent for Decision Tree. This means SVM was slightly more careful when assigning predicted classes, which can help reduce some false positive cases. However, SVM had lower Recall and F1-score than Decision Tree, so it was not selected as the strongest overall model in this comparison.

Overall, the final results support the use of machine learning for intrusion detection. The models were able to learn patterns from network traffic and classify the test records into different attack categories. The results also show why using more than one performance measure is important. If only precision was considered, SVM would look stronger. However, when Accuracy, Recall, and F1-score are also considered, Decision Tree becomes the better balanced choice for this dataset.

The feature selection analysis further showed that service related traffic attributes and error rate based features have strong relationships with the attack label. This means that intrusion detection systems can benefit from focusing on features that describe traffic behavior and host service patterns. The final model results are therefore consistent with the purpose of this project, which is to compare simple machine learning techniques and identify useful features for network intrusion detection.

7.0 Conclusion and recommendation

This project focused on developing a machine learning-based Network Intrusion Detection System using the NSL-KDD dataset. The main goal was to classify network traffic as either normal or attack traffic using two supervised machine learning models: Support Vector Machine (SVM) and Decision Tree. The models were implemented using Python in Google Colab and evaluated using Accuracy, Precision, Recall, and F1-score. Based on the experimental results, both models were able to detect intrusion patterns in the dataset. The Decision Tree model achieved the highest accuracy with 76.08%, while the SVM model achieved an accuracy of 75.31%. Although SVM achieved higher Precision, Decision Tree performed better in Accuracy, Recall, and F1-score. Therefore, Decision Tree was considered the better model in this project because it provided more balanced performance.

Several challenges were faced during the project, including understanding the NSL-KDD dataset, preprocessing categorical and numerical features, and dealing with the large dataset size. Another challenge was choosing suitable evaluation metrics, since Accuracy alone is not enough for intrusion detection. Precision, Recall, and F1-score were also used to better evaluate false alarms and missed attacks.

For future work, the system can be improved by testing more machine learning models such as Random Forest, Gradient Boosting, and Neural Networks. More advanced feature selection methods can also be applied to improve performance and reduce unnecessary features. In addition, using newer cybersecurity datasets such as CICIDS2017 or UNSW-NB15 may provide a more realistic evaluation of modern network attacks.

Overall, the results show that machine learning can support intrusion detection by identifying suspicious network traffic patterns. The project also shows that selecting a suitable model and using proper evaluation metrics are important for building an effective intrusion detection system.

The project team would like to thank Dr. Norah Alnaim for her guidance, support, and valuable feedback throughout this project. The team also appreciates the Computer Science Department at Imam Abdulrahman Bin Faisal University for providing the academic environment needed to complete this work.

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